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EXPERIMENTAL CHARACTERIZATION OF THE THERMOMECHANICAL BEHAVIOR OF POLYLACTIC ACID (PLA) SHAPE MEMORY POLYMERS

André Argôlo, andre.argolo@aluno.cefet-rj¹
Luã G. Gosta, lgcosta@mecanica.coppe.ufrj.br²
Vanderson M. Dornelas, vdornelas@id.uff.br³
Ricardo A.A. Aguiar, ricardo.aguiar@cefet-rj¹
Pedro M.C.L. Pacheco, pedro.pacheco@cefet-rj¹
Marcelo A. Savi, savi@mecanica.coppe.ufrj.br²

¹Department of Mechanical Engineering - Postgraduate Program in Mechanical Engineering and Materials Technology, CEFET/RJ – Centro Federal de Educação Tecnológica Celso Suckow da Fonseca, 20.271.110 Rio de Janeiro, RJ, Brazil

²Center for Nonlinear Mechanics, COPPE - Mechanical Engineering, Universidade Federal do Rio de Janeiro, 21.941.972 Rio de Janeiro, RJ, Brazil

³School of Metallurgical Industrial Engineering of Volta Redonda - Department of Mechanical Engineering, Universidade Federal Fluminense, 27.255.250, Volta Redonda, RJ, Brazil

Abstract: Polylactic acid (PLA)-based shape memory polymers (SMPs) exhibit thermomechanical behavior characterized by a thermally activated shape memory effect, enabling recovery of a programmed temporary shape upon heating above a characteristic glass transition temperature. Due to their biodegradability, lightweight nature, low cost, wide availability for manufacturing, and capacity to withstand high strains in specific conditions, PLA SMPs offer considerable versatility for applications in fields such as biomedicine, soft robotics, and deployable structures. Despite these advantages, their sensitive and complex thermomechanical response requires careful characterization for reliable use. This study investigates the thermomechanical properties of pure PLA fabricated via 3D printing. Differential scanning calorimetry (DSC) and tensile tests at different temperatures are conducted to evaluate its behavior and assess its potential as a functional shape memory material.

Keywords: Smart materials, Shape memory polymers, Polylactic acid, Thermomechanical behavior, 3D printing.

1. INTRODUCTION

Smart materials are characterized by their intrinsic capability to modify their physical state or behavior in response to external stimuli. This functionality arises from multiphysical couplings that integrate distinct physical domains, including mechanical, electrical, thermal, and magnetic interactions, among others. Within this broad class of materials, shape memory materials constitute a particularly notable subgroup.

Materials with the shape memory effect are able to undergo changes in their physical shape, usually used for strain recovery, when subjected to a specific external stimulus. In this context, shape memory alloys (SMAs) are the most widely known and studied materials. However, with the rapid development of advanced manufacturing techniques, particularly those associated with 3D printing processes, there has been a growing interest in shape memory polymers. These materials can recover from deformations similar to SMAs, but they have distinct characteristics that can be exploited in applications such as soft robots, biocompatible actuators, and others. Some of these applications are presented for Lendlein (2018) and Micalizzi *et al.* (2019).

Similarly to SMAs, shape memory polymers (SMPs) exhibit a strong sensitivity to temperature variations, particularly with respect to their mechanical behavior. Although both SMAs and SMPs present characteristic transition temperatures during heating up to the melting point, SMPs typically operate at significantly lower temperature ranges. Moreover, the mechanisms governing the shape memory effect in SMPs differ from those observed in SMAs. Xia (2021) provides a comprehensive overview of these materials, while Duarte (2023) investigates the influence of 3D printing parameters on the shape memory effect exhibited by them.

At low temperatures, SMPs generally behave as stiff and relatively brittle solids. As the temperature increases, molecular mobility rises, leading to a progressive reduction in stiffness accompanied by an increase in ductility. For a typical semi-crystalline thermoplastic, the elastic modulus remains relatively high and nearly constant at low temperatures. However, when the temperature exceeds a critical threshold, a pronounced drop in modulus occurs due to

the onset of significant molecular rearrangement. With continued heating, the material eventually reaches its melting point, at which it transitions into a viscous liquid state.

During heating, the material progresses through several distinct states, each associated with well-defined transition temperatures. Initially, the polymer is in a glassy state, characterized by higher rigidity and mechanical strength, but relatively low ductility. As the temperature increases, the material reaches the glass transition temperature (T_g), which experiences a phase transition from the glassy state to the rubbery state. For PLA, this transition typically occurs between 50 and 60 °C. With further heating, the material may reach the crystallization temperature, generally between 80 and 100 °C, followed by the melting temperature (T_m), which occurs in the range of approximately 155 °C to 170 °C. Among these thermal transitions, the shape memory effect primarily occurs within the glass transition region.

The shape memory effect in polymers occurs through a process commonly referred to as *programming*. In this process, the material is heated to a temperature within the glass transition (T_g) range to increase molecular mobility and ductility, allowing it to be deformed into a temporary shape. After deformation, the material is cooled while maintaining the imposed shape, locking it into a temporary configuration. During cooling, molecular mobility decreases significantly, preventing the immediate recovery of the deformation, and as a result, the applied stress is stored in the material in the form of elastic potential energy within the polymer network. Finally, when the material is reheated above the temperature at which the temporary shape is programmed, molecular mobility is restored, enabling the release of the stored energy and allowing the material to recover its original shape. It is important to note that this mechanism, like SMAs, has the capability for one-way and two-way behavior; however, the programming methods are more complicated and involve multiple temperatures for correct adjustment. Lendlein (2019) and Zhou *et al.* (2014) analyzed different approaches to enable one-way and two-way shape memory behavior in these materials. Additionally, multiple programmed shapes can be achieved using multiple materials, as demonstrated by Song *et al.* (2015).

The present work aims to provide an experimental characterization of PLA in order to establish a basis for the development of future numerical models. The study includes the analysis of the material transition temperatures through DSC tests, the evaluation of its mechanical behavior through rupture tests, and the investigation of the shape memory effect at different temperatures.

After this introduction, this paper is structured as follows. Section 2 describes the methodology and results from the DSC analysis. Section 3 presents the methodology and experimental results from tensile tests. Section 4 shows the shape memory effect tests. Finally, the conclusions of this study are summarized in Section 5.

2. DIFFERENTIAL SCANNING CALORIMETRY

One of the most important aspects when working with shape memory polymers is the determination of the material transition temperatures, since these parameters are essential for the programming process. For this reason, this section presents the DSC analysis performed to identify the transition temperatures of the tested material.

2.1. DSC Methodology

A Differential Scanning Calorimetry (DSC) test is performed to measure the material transition temperatures. The experiments are conducted using a *Netzsch DSC 200 F3 Maia* instrument, showed in Fig. 1(a). Two heating and cooling cycles are applied: the first cycle serves to eliminate any prior thermal history of the sample, while the second cycle is used to identify the characteristic transition temperatures. The tested material is the commercially available Creality's Hyper Series PLA, as depicted in Fig. 1(b), with a diameter of 1.75 mm, a recommended printing temperature range of 190–230 °C, and orange pigmentation. The tests are conducted over a temperature range from 20 °C to 200 °C, following the procedure described by Day (2006). As this maximum temperature exceeds the melting point by at least 30 °C, complete melting of the material is ensured, effectively eliminating its thermal history. The sample is maintained at 200 °C for 4 minutes, then cooled to 20 °C and held for an additional 4 minutes. This procedure is repeated once more, resulting in two heating cycles up to 200 °C and two cooling cycles down to 20 °C. A heating rate of 10 K/min is used, and the sample mass ranged from 5 mg to 10 mg.

The experiment enabled the accurate identification of the material thermal transition temperatures and the quantification of the enthalpic changes associated with each transition. These measurements capture the energy absorbed or released as the polymer evolves from the glass transition temperature (T_g) through subsequent thermal events up to the melting temperature (T_m), providing a comprehensive characterization of its thermomechanical behavior.

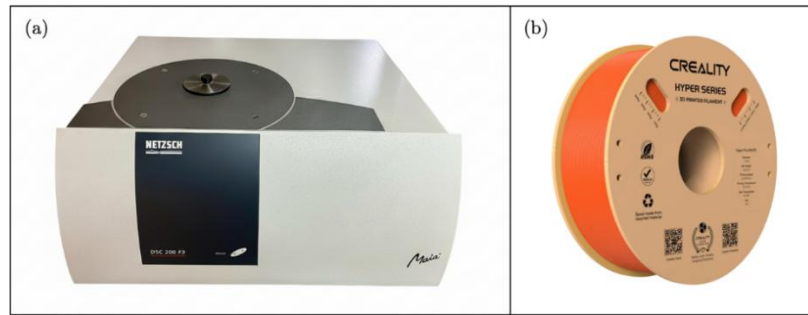


Figure 1. **Experimental approach for determining transition temperatures. (a) Differential Scanning Calorimeter (DSC 200 F3 Maia), used to measure the material's thermal transition temperatures; (b) Creality Hyper Series 3D-printing filament, from which the samples are obtained.**

2.2. DSC Results

For DSC analysis, samples are taken from a commercial PLA filament (see Fig. 1(b)). Figure 2 shows both the filament and the crucible used in the DSC measurements. The objective of these tests is to identify the characteristic phase transition temperatures and quantify the associated enthalpy variations, enabling a detailed characterization of the material thermal behavior. Three samples are analyzed, with masses of 5.5, 6.3, and 8.5 mg, respectively, being labeled according to the identification scheme presented in Tab. 1.



Figure 2. **PLA filament and corresponding DSC crucible employed in the thermal analysis.**

Table 1. Sample Identification.	
Sample Number	Sample Mass (mg)
A1	5.5
A2	6.3
A3	8.5

The DSC curves reveal the thermal transitions and corresponding energy variations for each tested sample. Based on the results, average values are computed to determine the representative transition temperatures and the associated energy changes during the phase transitions. Figure 3 presents the DSC curves for each tested sample, highlighting the glass transition (T_g), cold crystallization (T_c), and melting (T_m) regions. Table 2 summarizes the transition temperature ranges and the associated enthalpy values for each sample, along with their average values.

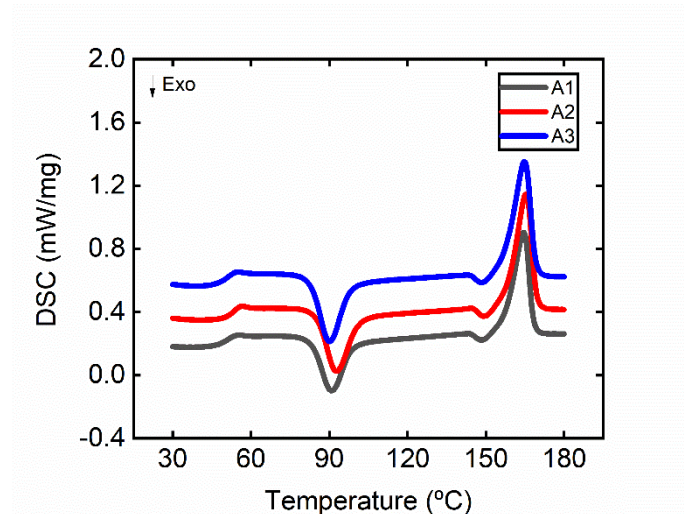


Figure 3. **DSC thermal analysis of PLA filaments, allowing the determination of the characteristic phase transition temperatures.**

Table 2. Characteristic phase transition temperatures and transition Energy of PLA filaments.

Sample Number	Transition Temperature Range (°C)			Transition Energy (J/g)		
	T_g	T_c	T_m	E_g	E_c	E_m
A1	46.70 - 59.10	83.30 - 98.60	157.30 - 168.30	0.49	22.01	30.96
A2	49.40 - 59.90	85.20 - 101.00	157.30 - 169.40	0.54	24.83	34.42
A3	46.90 - 59.10	82.90 - 97.80	156.90 - 169.00	0.65	24.74	37.86
Average Values	46.70 - 59.10	83.30 - 98.60	157.30 - 168.30	0.49	22.01	30.96

Both parameters are calculated directly using the DSC instrument software *Proteus Analysis* from Netzsch for DSC machines. To determine the transition temperatures, pairs of tangent lines are drawn at the onset and at the end of each phase transition. The intersection point of these tangents is defined as the corresponding transition temperature. The total energy associated with the transformation is obtained by calculating the area under the DSC curve.

3. TENSILE TESTS

The experimental analysis presented in this work is structured around two complementary approaches: the thermal characterization obtained via DSC testing, a mechanical response evaluation through rupture tests conducted at different temperatures, and the shape memory recovery with different temperatures within the transition temperature range.

3.1. Tensile Test Methodology

To characterize the mechanical behavior of the material, tensile tests are performed across a controlled temperature range to quantify the progressive reduction in tensile strength as the operating temperature approaches the glass transition temperature (T_g). This analysis is essential for assessing the potential degradation of structural integrity in components that may be exposed to elevated temperatures during service. The experiments are conducted using an *INSTRON 5982* universal testing machine, showed in Fig. 4(c), equipped with a temperature-controlled chamber that enables precise thermal conditioning of the specimens.

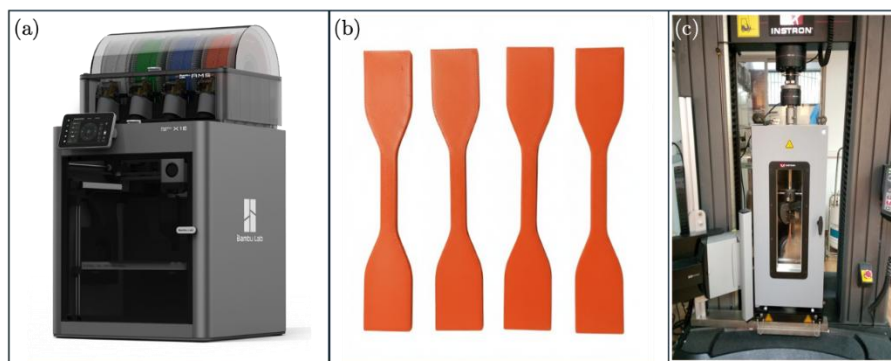


Figure 4. Equipment and specimens used in the tensile test: (a) Bambu Lab X1E 3D-printing machine, (b) printed PLA specimens, (c) universal testing machine INSTRON 5982.

The specimens are manufactured by 3D printing using a Bambu Lab X1E printer, showed in Fig. 4(a). The printing parameters included a 0.4 mm nozzle, 0.2 mm layer thickness, 45° raster angle, and 100% infill. The specimen geometry followed the ASTM D638-22 standard. Figure 4(b) presents the printed specimens utilized in testing.

For testing, the specimens are clamped using a manual wedge-action grip equipped with 1.0 mm flat serrated jaw faces. A length of 25mm at each end of the specimen is secured within the grips, resulting in an effective gauge length of 65 mm from a total specimen length of 115 mm. All tests are performed at a constant prescribed displacement rate of 5 mm/min.

3.2. Failure Test Results

For the tensile tests, a preliminary failure assessment is conducted at temperatures ranging from 25 °C to values approaching the onset of the glass transition. The purpose is to quantify the reduction in mechanical performance as temperature increases, capturing the reduction of strength and stiffness before the transition region. Based on the DSC results, which indicate a T_g onset at approximately 47 °C, specimens are tested at 25, 35, and 45 °C. Figure 5 presents the corresponding stress-strain curves, while Tab. 3 reports the yield stress (MPa), ultimate stress (MPa), and maximum elongation (%) for each specimen.

Table 3. Rupture Test Results

Specimens	Yield stress (MPa)	Rupture stress (MPa)	Maximum elongation (%)
R1	38.4	21.4	7,9%
R2	39.4	23.1	134.9%
R3	23.8	21.3	226.4%

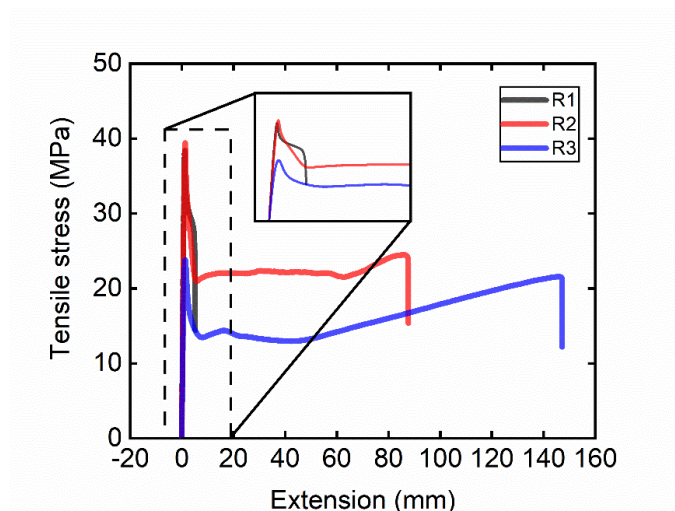


Figure 5. Failure analysis performed at different temperatures. R1(Black): 25 °C; R2(Red): 35 °C; R3(Blue): 45 °C.

At 25 °C, the stress–strain response exhibits the typical behavior expected for PLA under ambient conditions, characterized by a well-defined elastic region followed by brittle fracture. The material reaches a yield stress of approximately 40 MPa and fails at a strain of 7.9% (≈ 5 mm displacement).

At 35 °C, the overall curve shape remains similar to that observed at 25 °C, and yield stress remains basically unchanged; however, the maximum deformation increases to around 90 mm, corresponding to a strain of 134.9%. This indicates a significant increase in ductility without a noticeable variation in mechanical strength. In this temperature range, the deformation capacity increases considerably, while the yield stress varies only within a negligible margin.

At 45 °C, the mechanical response changes noticeably. The yield stress decreases substantially to approximately 25 MPa, reflecting the onset of thermal softening as the temperature approaches the glass transition region. Concurrently, the strain at failure increases considerably, reaching approximately 150 mm (around 226.4% strain), indicating a transition toward a highly ductile behavior. This combination of reduced strength and increased deformability is consistent with the proximity to the glass transition temperature identified in the DSC analysis.

These results highlight the strong temperature dependence of PLA, demonstrating that small increases in temperature near the glass transition can significantly alter its mechanical response. This behavior indicates the potential for using the material in applications that require a combination of mechanical strength and high ductility, with only minor adjustments to the operating temperature.

4. SHAPE MEMORY EFFECT

Shape memory polymer effect is obtained by a programming process that, as previously discussed, involves controlled heating and cooling cycles at specific temperatures. This section presents the methodology and the experimental results related to the shape memory effect exhibited by pure PLA.

4.1. Shape Memory Effect Cycle Methodology

The test sequence consisted of heating the specimen to the programming temperature, deforming it to a prescribed strain, cooling it to room temperature, and subsequently unloading it until the applied stress reached a near-zero value. The specimen is then reheated to promote shape recovery, while the testing machine maintains the prescribed stress approximately constant throughout the recovery process.

A strain level of 6% is applied to the specimens at a loading rate of $1\% \cdot \text{min}^{-1}$ to determine the maximum tensile stress, and to quantify the recovery time required to recover 60% of the imposed deformation. This recovery limit is primarily selected to restrict the total test duration to a maximum of one hour.

4.2. Shape Memory Effect Cycle Results

The shape memory effect analysis is performed by programming and recovering the material deformation at four initial temperatures: 45 °C, 50 °C, 55 °C, and 60 °C, all within the transition range identified during the DSC analysis. However, during the experiments, it was observed that the tests conducted at 55 °C and 60 °C did not exhibit significant differences in behavior. Therefore, the analysis is subsequently limited to the temperatures of 45 °C, 50 °C, and 55 °C. It is important to emphasize that, for the 55 °C cycle, the loading rate is increased from the initially adopted value of 1%/min to 2%/min. For the prescribed stress, a 5N load is held constantly throughout the recovery process for the temperatures of 45°C and 50°C, while for 55°C, the load held during the recovery process is 2N.

This adjustment is made because a significant amount of noise is observed during loading, which is believed to be associated with the extremely low stress levels. Under these conditions, variations that would normally be negligible became more pronounced in the experimental response. Therefore, the loading rate is increased in order to obtain smoother and cleaner loading curves with reduced noise influence.

Figure 6 presents the shape memory cycles obtained for the three selected temperatures. From these curves, it can be observed that, at the lowest temperature, the material exhibits a considerably higher stress level compared to the other tested temperatures. As expected, the response even reaches the yielding region, which can be clearly identified in the graph.

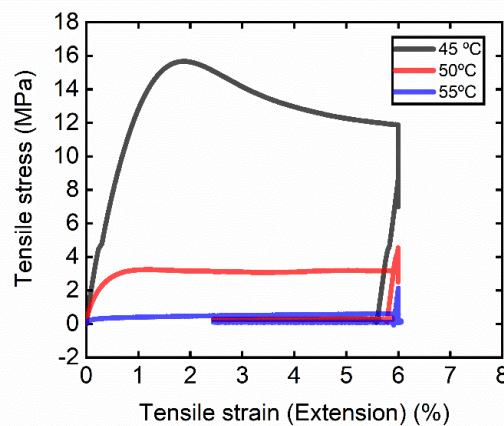


Figure 6. Shape memory effect cycles for the three temperatures.

At 50 °C, a substantially lower stress level is observed; however, the material still presents a noticeable resistance to deformation. In contrast, at 55 °C, the material exhibits virtually no resistance to deformation, with extremely low stress levels being required to achieve the same imposed strain. However, it is also important to evaluate the recovery time of the material at each tested temperature. As previously mentioned, the cycle performed at 55 °C used a higher loading rate than the other tests, resulting in an earlier onset of deformation recovery. To adjust for that, the recovery time is measured individually for each test, starting from the stabilization of the prescribed stress until the target recovery level is achieved. Figure 7 presents the evolution of the applied deformation as a function of time for each tested temperature.

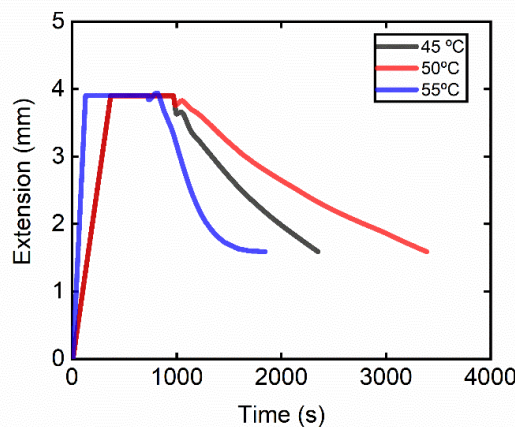


Figure 7. Evolution of the extension over time.

From Figure 7, it can be observed that there is a significant difference in recovery time between the tests performed at 45 °C and 50 °C, with the specimen tested at 50 °C exhibiting a considerably longer recovery time. In contrast, after accounting for the different loading conditions, the test conducted at 55 °C presented the shortest recovery time among the three evaluated temperatures. Table 4 summarizes the recovery times obtained for each test.

Table 4. Recovery time for each tested temperature.

Test Temperature	Recovery Time (Minutes)
45°C	22,46
50°C	38,46
55°C	18,51

5. CONCLUSIONS

This work presents a preliminary thermomechanical characterization of pure 3D printed polylactide acid (PLA) shape memory polymers. The characterization is performed in three phases: the thermal characterization of the raw material via Differential Scanning Calorimeter (DSC), which enables the identification of the glass transition, cold crystallization, and melting temperature regions, the mechanical characterization of 3D printed specimens through tensile tests performed at different temperatures and the characterization of the shape memory effect which is tested within the temperature range where the glass transition occurred.

The results demonstrate that the mechanical response of PLA is highly sensitive to temperature variations, particularly as the material approaches the glass transition region. A progressive reduction in strength, accompanied by a significant increase in ductility, is observed with increasing temperature, highlighting the strong coupling between thermal and mechanical behavior. Additionally, the recovery rate of the shape memory effect can vary considerably within the glass transition range; however, it is important to note that the recovery time observed in this test is considerably longer than what is typically expected for shape memory polymers. This behavior is mainly associated with the experimental setup, since the test was performed using a tensile testing machine to impose the deformation on the specimen. For future investigations, it would be beneficial to design an experimental setup capable of applying tensile loading while simultaneously allowing the material to recover freely during the recovery stage.

Future work will also investigate potential shifts in phase transition temperatures associated with the pigmentation additives. In addition, the shape-memory behavior of the material will be examined in more detail, with particular emphasis on hysteresis effects during the recovery cycle.

6. ACKNOWLEDGMENTS

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